

Then the operator $T(u) \equiv f(u) - P_{\Phi(u)}(f(u) - \alpha u)$ is Lipschit continuous.

Proof. By the non-expansion property of the projection operator and (2.6) we have that

$$\begin{aligned} \|T(u) - T(v)\| &= \|f(u) - P_{\Phi(u)}(f(u) - \alpha u) - (f(v) - P_{\Phi(v)}(f(v) - \alpha v))\| \\ &\leq \|f(u) - f(v)\| \\ &\quad + \|P_{\Phi(u)}(f(u) - \alpha u) - P_{\Phi(u)}(f(v) - \alpha v) + P_{\Phi(u)}(f(v) - \alpha v) - P_{\Phi(v)}(f(v) - \alpha v)\| \\ &\leq L\|u - v\| + \|f(u) - f(v)\| + \alpha\|u - v\| + \mu\|u - v\| \\ &\leq (2L + \alpha + \mu)\|u - v\|. \end{aligned}$$

This implies that T is Lipschit continuous. $\square$

3. FINITE-TIME STABILITY ANALYSIS

In this section we will analyze the finite-time stability for IQVIPs (1.1). By doing so we propose a new first order dynamical system associated with the problem (1.1) such that a solution of (1.1) becomes an equilibrium point of the dynamical system. Under mild conditions for the parameters, we demonstrate that the proposed dynamical system is finite-time stable.

We first recall a characterization for the finite-time stability of solutions of a dynamical system, given by Bhat and Bernstein in [5]. This result will play an important role in studying finite-time convergence.

Theorem 3.1. [5] (*Lyapunov condition for finite-time stability*). Assume that there exists a continuously differentiable function $V : D \rightarrow \mathbb{R}$, where $D \subseteq \mathbb{R}^n$ is a neighborhood of the equilibrium point u^* for (2.1) and an open neighborhood $U \subseteq D$ of u^* such that

$$\dot{V}(u) \leq -K \cdot (V(u))^p \quad \forall u \in U \setminus \{u^*\},$$

where $K > 0$ and $p \in (0, 1)$. Then, the equilibrium point u^* of (2.1) is finite-time stable equilibrium point of (2.1). Moreover, the settling time T satisfies

$$T(u(0)) \leq \frac{V(u(0))^{1-p}}{K(1-p)}$$

for any $u(0) \in U$.

In addition, if $D = \mathbb{R}^n$, then the equilibrium point u^* of (2.1) is globally finite-time stable.

In order to obtain a finite-time stability of solutions of inverse quasi-variational inequality problem (1.1) we now present a novel first-order dynamical system associated with the problem (1.1). Our proposed dynamical system is as follows:

$$\dot{u}(t) = -\sigma \frac{f(u) - P_{\Phi(u)}(f(u) - \alpha u)}{\|f(u) - P_{\Phi(u)}(f(u) - \alpha u)\|^{\frac{\gamma-2}{\gamma-1}}}, \quad (3.1)$$

where $\sigma > 0$ is a scalar tuning gain, and $\gamma > 2, \alpha > 0$ are two design parameters.

It is worth noting that for all $u \in \mathbb{R}^n$ such that $f(u) - P_{\Phi(u)}(f(u) - \alpha u) = 0$, the expression on the right-hand side of (3.1) is still well defined for all $\gamma > 2$. In fact, it is equal to zero. Indeed, we have

$$\left\| \frac{f(u) - P_{\Phi(u)}(f(u) - \alpha u)}{\|f(u) - P_{\Phi(u)}(f(u) - \alpha u)\|^{\frac{\gamma-2}{\gamma-1}}} \right\| = \|f(u) - P_{\Phi(u)}(f(u) - \alpha u)\|^{\frac{1}{\gamma-1}} = 0.$$